# HOMEWORK ASSIGNMENT 9

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

### QUESTION 1: Pool Party!

If you've ever flown across California in a window seat, you might have noticed a speckling of pools. But do different California cities differ in the prevalence of homes that have pools? Let's make this more specific: does the population proportion of houses for sale that have a pool differ across mid-sized coastal California cities?

To answer this question, I (Dr. Ethan) randomly selected six mid-sized coastal cities: two from northern California (Vallejo and Redding), two from the central coast (Santa Maria and Ventura), and two from southern California (Costa Mesa and Carlsbad). I then went to *Zillow*[1] and searched for houses for sale; I then recorded the total number of houses in each of the six cities and the number of those houses that have a pool. Here is a summary of those counts:

| City | Redding | Vallejo | Santa Maria | Ventura | Costa Mesa | Carlsbad |
|---|---|---|---|---|---|---|
| **With Pool** | 71 | 11 | 4 | 7 | 19 | 19 |
| **Total** | 361 | 246 | 141 | 115 | 93 | 122 |

**Table 1:** Summary of the Data

Whenever necessary we'll consider "having a pool" to be a success.

(a) Describe an observational unit in the context of this study.

(b) Identify the explanatory and response variables, and classify each as either categorical or quantitative. For any categorical variables, clearly specify the levels.

(c) Was random sampling utilized? What about random assignment? Explain.

(d) State the null and alternative hypotheses in words and, if possible, symbols and math.

(e) Calculate the conditional proportions.

(f) Fill in Table (2) at the top of the next page to calculate the mean group difference statistic.

**Action Item 1**

We'll now let the Multiple Proportions applet do some work for us.

**1)** Click Here to access the data; select-all and copy.

**2)** Navigate to the Multiple Proportions applet (if the link above is not working, use the steps outlined in Lecture Activity 18).

**3)** Click into the "**Paste data or table below**" box; select-all, and paste the data.

**4)** Click "**Use Table**" (**not** "Use Data").

[1] https://www.zillow.com/

| Groups Diff. | Difference in Sample Proportions | Absolute Value | Mean Group Diff. |
|---|---|---|---|
| Redding – Vallejo | | | |
| Redding – Santa Maria | | | |
| Redding – Ventura | | | |
| Redding – Costa Mesa | | | |
| Redding – Carlsbad | | | |
| Vallejo – Santa Maria | | | |
| Vallejo – Ventura | | | |
| Vallejo – Costa Mesa | | | |
| Vallejo – Carlsbad | | | |
| Santa Maria – Ventura | | | |
| Santa Maria – Costa Mesa | | | |
| Santa Maria – Carlsbad | | | |
| Ventura – Costa Mesa | | | |
| Ventura – Carlsbad | | | |
| Costa Mesa – Carlsbad | | | |

**Table 2:** Calculating the Mean Group Diff

(g) Which will be more informative: a segmented bargraph, or a mosaic plot? Use the applet to create the plot you think is more informative, and include a screenshot.

(h) Is the mean group difference statistic value reported by the applet similar to the one you obtained in part (f) above? If it is slightly different, what do you think the reason for that is?

(i) Calculate and report the approximate simulation-based $p$-value, using at least 10,000 simulated shuffles. Then, both *interpret* and **evaluate** this $p$-value.

(j) Formulate a complete conclusion in the context of the study. Make sure, in addition to including a clear statement about the parameters, you address:

- Significance: was there a statistically significant effect or not?
- Generalization: to what population can we extend our results?
- Causation: can we make a causal claim? If so, what is the claim?

(k) Hopefully you found that the data leads us to reject the null hypothesis in favor of the alternative, indicating that there *is* a difference in the prevalence of pools between the six cities included in the

study. A natural follow-up question is: "which cities are different?" We know that one way to address this question is by constructing **pairwise confidence intervals**.

i. Use the applet to construct all 15 pairwise confidence intervals, and identify which pairs of cities exhibit statistically discernible differences in population proportions of for-sale houses that have pools.

ii. Pick one of the intervals that indicate a significant result, and interpret it **in context**.

### QUESTION 2: Aced It

Do different professors give out different proportions of "A" grades, even across the same class? Jasleny, a student at a fictional university, decides to focus her attention on BIO 101, a large introductory biology course. Last semester (Spring 2026), this course had four lecture sections, each taught by a different professor (who we will call "1," "2," and "3"). Using publicly-available data, Jasleny calculated the number of BIO 101 students each professor awarded an "A" last semester, and produced the following table:

| | **Prof 1** | **Prof 2** | **Prof 3** | **Prof 4** | **Total** |
|---|---|---|---|---|---|
| **A** | 15 | 19 | 13 | 14 | 61 |
| **Not A** | 85 | 81 | 87 | 86 | 339 |
| **Total** | 100 | 100 | 100 | 100 | 400 |

**Table 3:** Summary of the Data

(a) State the null and alternative hypotheses in symbols and math only, if possible. If it is not possible to write a hypothesis in symbols and math only, write it as a verbal statement that involves the parameter(s) of interest. **Note:** I do not want the hypotheses in terms of "no association" and "association."

(b) Calculate the value of the $\chi^2$ statistic. Show your work!

(c) Explain why the validity conditions are satisfied to proceed with a theory-based approach.

**Action Item 2**

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**4)** Click "**Use Table**" (**not** "Use Data").

(d) Calculate a simulation-based *p*-value, using the $\chi^2$ statistic and at least 10,000 repetitions.

(e) Calculate the theory-based *p*-value, using the $\chi^2$ statistic.

(f) Given the results of your analysis, provide an overall conclusion **in context**. Please include a **generalization** statement. Refer to the parameter(s) and be very clear what you can claim about the population proportions.

(g) Is it appropriate to examine the confidence intervals for the pairwise differences? If so, explain why, and report and interpret the intervals that indicate a significant difference. If not, explain why examining the confidence intervals is not appropriate.

(h) Are we justified in making a causal claim?

- ◯ This is an observational study; therefore, we cannot make any causal claims because of the presence of confounding variables.
- ◯ This is an observational study; therefore, we can make any causal claims because confounding variables are balanced out between the groups.
- ◯ This is a randomized experiment; therefore, we can make any causal claims because confounding variables are balanced out between the groups.
- ◯ This is a randomized experiment; therefore, we cannot make any causal claims because of the presence of confounding variables.